
1 RELATED WORK

1.1 AGENT MEMORY SYSTEMS

Recent surveys argue that agent memory is evolving into a full-fledged data management layer that governs storage, retrieval, consolidation, and lifecycle policies throughout execution (Zhou et al., 2026). Complementing this view, enterprise-grade substrates emphasize durable, organization-wide retention of task state and procedural knowledge for long-horizon workflows (Alake et al., 2026). At the mechanism level, memory-as-a-tool approaches convert transient critiques into retrievable guidelines to amortize inference-time reasoning (Gallego, 2026). Unified episodic–semantic designs further explore spreading-activation retrieval to connect long-term facts with task-specific episodes (?).

1.2 RETRIEVAL METHODS FOR AGENT MEMORY

Neural ranking and retrieval provide the algorithmic backbone for scoring and selecting relevant records under diverse evidence sources (Trabelsi et al., 2021). Hybrid retrievers that fuse vector semantic search with lexical signals have demonstrated accuracy gains over single-source methods (Sawarkar et al., 2024; ?). Beyond query–content similarity, longitudinal studies highlight the importance of temporal dynamics in retrieval effectiveness (Ovcharov, 2026). Our setting adopts a hybrid scoring paradigm tailored to agent memory by integrating semantic similarity with temporal decay, access frequency, and importance weighting.

1.3 MULTI-AGENT COLLABORATION

Multi-agent evaluation frameworks underscore the benefits and challenges of coordinating several LLM agents to assess capabilities and systemic risks (Rasheed et al., 2024). Domain-specific, hierarchical teams showcase how role specialization and delegated reasoning can drive autonomous scientific discovery (Rothfarb et al., 2025). From a systems perspective, organization-scale memory infrastructures are advocated to support coherent collaboration across extended processes (Alake et al., 2026). Nevertheless, concrete protocols for synchronizing, reconciling, and governing shared memories across teams remain under-specified.

1.4 LLM-BASED AUTONOMOUS AGENTS

LLM-based agents increasingly rely on tool use and iterative interaction to execute complex tasks, yet alignment and finetuning must bridge the gap between generation and effective action (Wang et al., 2024). Surveys of advanced LLM capabilities highlight persistent issues—temporal knowledge gaps, arithmetic robustness, and hallucination—that motivate external memory and retrieval (Roffo, 2024). As agents accumulate long-term histories, privacy analyses reveal new attack surfaces in memory stores, calling for principled management and access policies (Chen et al., 2026). In parallel, techniques that distill feedback into persistent, retrievable rules aim to stabilize behavior across episodes (Gallego, 2026).

Building on calls for agent-native memory infrastructures, we introduce a three-level working–task–project architecture with explicit per-layer TTL and capacity controls, enabling lifecycle governance that is not addressed by existing designs (Zhou et al., 2026). Departing from standard hybrid retrievers, our scoring function couples semantic similarity with temporal decay, access frequency, and importance weighting to prioritize salient, timely, and reused items (Sawarkar et al., 2024; Ovcharov, 2026). Finally, unlike prior multi-agent efforts that emphasize evaluation or hierarchical role decomposition, we formalize a team memory synchronization protocol and validate the full stack in JiuwenSwarm (Rasheed et al., 2024; Rothfarb et al., 2025).

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